

1. In 2022, the Negro Leagues Baseball Silver Dollar was released by the U.S. Mint as part of the Negro Leagues Baseball Commemorative Coin Program. The faces of the coin are shown.

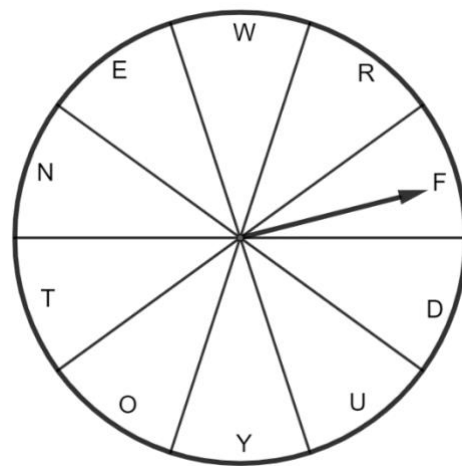


For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
a. Flipping the Negro Leagues Silver Dollar	{Pitcher, Batter}
b. Rolling a fair 17 – sided number cube 	{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17}

2. In a game, players spin a fair spinner to determine a letter that they must use. The spinner has 7 consonants and 3 vowels on it.

Find the theoretical probability of each event. Write the probability as a percentage.



	Event	Theoretical Probability
a.	Spinner lands on the letter "T".	10%
b.	Spinner lands on a consonant.	70%

3. Complete the table by writing the equivalent values in each form.

	Fraction	Decimal	Percent
a.	$\frac{6}{11}$	$0.\overline{54}$	$54.\overline{54}\%$
b.	$\frac{85}{100}$ or equivalent	0.85	85%
c.	$\frac{72}{100}$ or equivalent	0.72	72%

4. For each context, place an **X** in the box for all of the simulations that could be used to simulate the probability.

	Simulations		
	A deck of 16 cards with 16 different letters.	A fair spinner with 7 equal sections where each is a different color.	A 10-sided die with the numbers 1 through 10.
A baseball player with an on-base percentage of 70%.	☐	☐	☒
A basketball player making $\frac{5}{7}$ free throws.	☐	☒	☐
A soccer goalie who stops $\frac{1}{2}$ of shots on the goal.	☒	☐	☒
A hockey player with a shooting percentage of 25%.	☒	☐	☐

5. Complete the statement.

To use the

- ☐ 10-sided die
- ☐ deck of 16 cards
- ☒ fair spinner with 7 equal sections

to simulate the

- ☐ soccer
- ☐ hockey
- ☐ baseball
- ☒ basketball

context, 5

- ☐ cards
- ☐ numbers
- ☒ equal sections

should be assigned to

- ☐ being on base.
- ☒ making a free throw.
- ☐ stopping a shot on goal.
- ☐ shooting a goal in hockey.

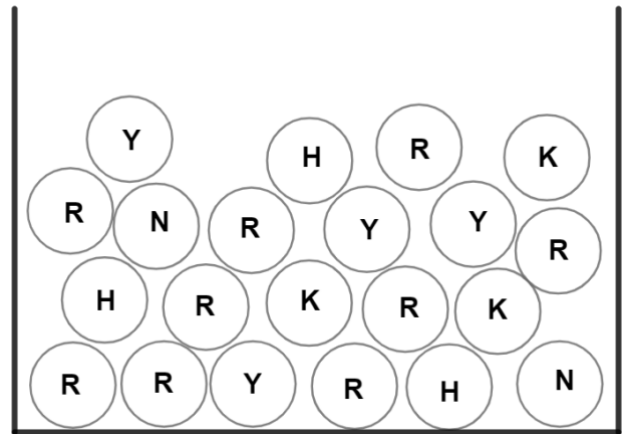
***Note:** There are two correct ways to complete this statement. The answer shown

OR

10-sided die; soccer; numbers; stopping a shot on goal.

6. A jar with 21 marbles is shown.

Determine each theoretical probability. Write the answer in fraction form.



	Event	Theoretical Probability
a.	Drawing a marble with the letter R or Y from the jar	$\frac{13}{21}$
b.	Drawing a marble with a letter other than H from the jar	$\frac{18}{21}$ or $\frac{6}{7}$
c.	Drawing a marble with the letter N from the jar	$\frac{2}{21}$

d. Complete the statements.

The event that is less likely to occur is

- drawing a marble with the letter N from the jar.
- drawing a marble with the letter R or K from the jar.
- drawing a marble with a letter other than H from the jar.

The event that is most likely to occur is

- drawing a marble with the letter N from the jar.
- drawing a marble with the letter R or K from the jar.
- drawing a marble with a letter other than H from the jar.

The probability of the less likely event is closer to

- 0.
- 1.

7. Aaliyah and Xavier did a computer simulation for drawing a marble with the letter R from the jar of 21 marbles. Each used a spinner with 21 equal sections where 3 sections are labeled H, 9 sections are labeled R, 3 sections are labeled K, 4 sections are labeled Y, and 2 sections are labeled N. The number of spins and the results are shown in the table.

	Aaliyah	Xavier
Spins	42	21,000
Results	$\frac{10}{42}$	$\frac{8982}{21000}$

The number of spins in

- ☐ Aaliyah's
- ☒ Xavier's

simulation results in a(n)

- ☒ experimental
- ☐ theoretical

probability that is closer to the

- ☐ experimental
- ☒ theoretical

probability.

8. The probabilities of four events are shown.

75%

9 out of 10

$\frac{8}{11}$

0.77

List the probabilities in order from least to greatest.

Least to Greatest

$\frac{8}{11}$, 75%, 0.77, 9 out of 10

9. A package of Tropical Skittles has five flavors: Banana Berry (yellow), Kiwi Lime (green), Mango Tangelo (orange), Pineapple Passion fruit (blue), and Strawberry Starfruit (pink). The table shows the distribution of each flavor in a randomly chosen package.

Flavor	Amount
Banana Berry	20%
Kiwi Lime	$\frac{4}{15}$
Mango Tangelo	0.12
Pineapple Passion fruit	$19.\bar{1}\%$
Strawberry Starfruit	$0.\bar{2}$

- a. What percentage of Maria's package is Mango Tangelo?

Percentage	12%
-------------------	-----

- b. Rewrite the amount of Maria's package that is Kiwi Lime as a decimal.

Decimal	$0.2\bar{6}$
----------------	--------------

- c. What fraction of Maria's package is Strawberry Starfruit?

Fraction	$\frac{2}{9}$
-----------------	---------------

- d. Rewrite the amount of Maria's package that is Pineapple Passion fruit as a fraction.

Fraction	$\frac{43}{225}$ or equivalent fraction
-----------------	---

- e. Rewrite the amount of Maria's package that is Banana Berry as a decimal.

Decimal	0.2
----------------	-----